Learned Molecular Graph Representations versus Engineered Fingerprints for Multitask Toxicity Prediction on Tox21

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***Abstract*—Toxicity screening of chemical libraries is slow and costly, motivating models that predict assay activity directly from molecular structure. This paper asks whether representations *learned* by graph neural networks outperform *engineered* chemical fingerprints on Tox21, a benchmark of 7,831 compounds and 12 toxicity endpoints. After canonicalisation, salt stripping and deduplication (7,686 unique molecules), we train three message passing architectures — a graph convolutional network, a bond-aware GINE network and AttentiveFP — as multitask models over all twelve endpoints using a masked, class-weighted cross-entropy that ignores the 17% of labels that are missing. A deep multitask perceptron on the same fingerprints isolates the effect of deep learning from that of the graph representation, and two hybrids combine both views. All models are compared on three Bemis–Murcko scaffold splits, which hold out entire chemical families, using endpoint-wise ROC-AUC and PR-AUC computed on observed labels only and paired Wilcoxon tests. On unseen scaffolds the graph networks reach 0.803–0.818 ROC-AUC and are statistically indistinguishable from gradient boosting (0.816) and from the fingerprint perceptron (0.814), while a random forest over fingerprints and descriptors remains significantly stronger (0.834). Graph models also lose roughly twice as much accuracy as tree ensembles when moving from random to scaffold splits. The two views are nevertheless complementary: a validation-weighted late fusion of a random forest and a graph network attains the best overall result (0.844 ROC-AUC, 0.503 PR-AUC). At this data scale, learned graph representations do not replace engineered descriptors, but they contribute information that improves prediction when the two are combined.**

***Index Terms*—Graph neural networks, message passing, molecular property prediction, multitask learning, missing labels, scaffold split, Tox21, toxicity prediction.**

I. INTRODUCTION

Regulatory agencies and pharmaceutical companies must assess the toxicity of tens of thousands of chemicals, yet every *in vitro* or *in vivo* assay is slow, expensive and, in the animal case, ethically costly. Models that predict assay outcomes from chemical structure alone can rank untested compounds, flag likely hazards early and reduce experimental burden. Their practical value depends almost entirely on how well they extrapolate to *new* chemistry, because a screening model is applied precisely to compounds that differ from those used to build it.

Two families of molecular representation dominate this task. *Engineered fingerprints*, such as Morgan or extended-connectivity fingerprints, hash local atom neighbourhoods into a fixed-length bit vector and are paired with classical learners such as random forests or gradient boosting. They are fast, robust and remarkably hard to beat, but they are fixed: the same substructure dictionary serves every endpoint, bit collisions discard information, and nothing in the representation adapts to the prediction task. *Graph neural networks* (GNNs) instead treat a molecule as a graph of atoms and bonds and *learn* a task-specific representation by message passing. In principle they can capture interactions that a fixed dictionary cannot, and a multitask network can share chemistry across biologically related assays.

In an earlier phase of this project we established the classical baseline: under a scaffold split, a random forest over Morgan fingerprints and RDKit descriptors reached 0.834 ROC-AUC, ahead of gradient boosting (0.816) and logistic regression (0.792). This paper reports the deep learning phase and answers the question that motivated it:

> *Do learned molecular-graph representations outperform engineered fingerprints for toxicity prediction, and how does each generalise to unseen chemical families?*

The contributions of this phase are as follows.

- Three structurally different message passing architectures — GCN, bond-aware GINE and AttentiveFP — trained as multitask models over all twelve endpoints with a *masked*, class-weighted loss, so that the 17% of labels that are missing contribute neither loss nor gradient.
- Controls that isolate the source of any difference: a deep multitask perceptron over exactly the engineered features used by the baselines, and a descriptor ablation.
- Two ways of combining the views — an end-to-end hybrid network and a validation-weighted late-fusion ensemble — together with an analysis of why joint training is harder than it appears.
- A generalisation study reporting the random-versus-scaffold performance gap per model and endpoint, and accuracy as a function of each test molecule's similarity to the training set.
- A statistically grounded comparison using paired Wilcoxon signed-rank tests over 36 endpoint–seed pairs, with a fully reproducible implementation.

II. RELATED WORK

Benchmarks. The Tox21 Data Challenge [1] released twelve nuclear-receptor and stress-response assays screened over roughly 10,000 environmental chemicals and drugs. MoleculeNet [2] standardised Tox21 as a multitask benchmark, showed that graph convolutions were competitive with fingerprint models under random splits, and recommended scaffold splitting as a more realistic estimate of generalisation.

Engineered representations. Extended-connectivity fingerprints [3] iteratively hash circular atom environments into a bit

vector and remain a strong default for activity prediction, typically combined with random forests [4] or gradient boosted trees [5]. Bemis and Murcko [6] introduced the molecular framework, or scaffold, that underlies scaffold-based data splitting.

Message passing networks. Gilmer *et al.* [7] unified molecular graph networks under the message passing framework. Graph convolutional networks [8] aggregate neighbours with degree normalisation; the graph isomorphism network [9] uses sum aggregation and is provably as discriminative as the Weisfeiler–Lehman test, and its edge-aware variant GINE [10] injects bond features into each message. AttentiveFP [11] applies graph attention together with a gated readout and reported strong Tox21 results. Yang *et al.* [12] found with Chemprop that learned representations surpass fingerprints on many but not all datasets, that concatenating global RDKit descriptors helps considerably, and that scaffold splits depress every model's apparent performance.

Our study follows this line but focuses on a controlled, statistically tested comparison under scaffold splitting, with explicit controls separating the contribution of deep multitask learning from that of the graph representation itself.

III. DATASET AND PREPROCESSING

We use the MoleculeNet/DeepChem release of Tox21: 7,831 SMILES strings with binary labels for twelve endpoints, seven nuclear-receptor (NR) assays and five stress-response (SR) assays. A label is missing when the compound was not tested in that assay or the result was inconclusive.

A. Cleaning and deduplication

Every SMILES is parsed with RDKit, standardised, reduced to its largest organic fragment so that salts and counter-ions are stripped, and written as a canonical SMILES. Rows are then grouped by canonical SMILES so that a parent compound has a single identity. For each endpoint, duplicate labels are merged when they agree and set to missing when they conflict. Crucially, deduplication happens *before* any split, so the same compound can never appear in both training and test data. Table I summarises the resulting data accounting.

TABLE I
Data Cleaning Summary

Stage	Count
Raw rows	7,831
Unparseable SMILES removed	8
Duplicate rows merged	137
Conflicting labels set to missing	60
Final unique molecules	7,686
Observed labels	76,536 (83.0%)
Molecules with all twelve labels	3,019 (39%)
Unique Bemis–Murcko scaffolds	2,280

B. Class imbalance and missing labels

The endpoints are severely imbalanced: the active rate ranges from 2.8% (NR-PPAR-gamma) to 15.9% (SR-ARE). They are also incomplete: 17% of labels are missing and only 3,019 molecules (39%) carry all twelve labels. Discarding incomplete molecules would therefore remove the majority of the data, while imputing missing entries as inactive would inject false negatives into assays where the compound was simply never tested. Both problems are handled in the loss rather than in the data, as described in Section IV-D.

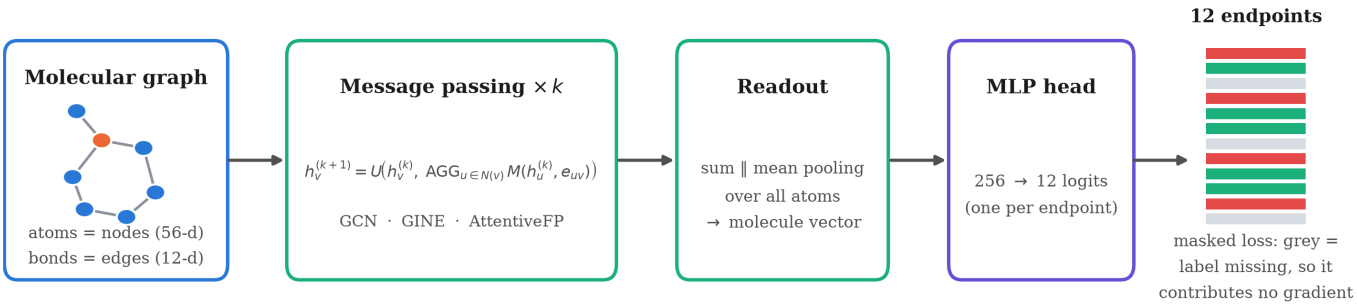


Fig. 1. Multitask message passing pipeline. A molecule is encoded as a graph with 56-dimensional atom features and 12-dimensional bond features. After k rounds of message passing, a permutation-invariant readout produces a single molecule vector and an MLP head emits twelve logits. Endpoints whose label is missing (grey) are masked out of the loss and contribute no gradient.

IV. METHODOLOGY

A. Scaffold splitting

Every model is evaluated on identical, cached split files. The primary protocol is a randomised (balanced) Bemis–Murcko scaffold split in 80/10/10 proportions, giving 6,148 training, 768 validation and 770 test molecules. Each scaffold group is assigned to exactly one partition, so test molecules belong to chemical series entirely absent from training. Scaffold groups larger than half the test set — notably the acyclic group (1,739 molecules) and the benzene group (1,476), each larger than the whole test set — are placed in training first, and the remaining groups are shuffled with the random seed. Seeds 0, 1 and 2 produce three independent scaffold splits in which no scaffold is shared between training and test. Three molecule-level random splits of identical size are used only to quantify the generalisation gap.

B. Graph representation

Atoms are nodes and each bond contributes two directed edges. Atom features (56 dimensions) comprise one-hot encodings of element (22 common elements plus an "other" slot), total degree, formal charge, hybridisation, attached hydrogen count and chirality tag, together with aromaticity, ring membership and scaled atomic mass. Bond features (12 dimensions) encode bond type, stereochemistry, conjugation and ring membership.

C. Architectures

Fig. 1 summarises the pipeline shared by all three graph models. All networks are multitask, emitting twelve logits. In a message passing layer each atom v updates its hidden state from its neighbourhood $N(v)$ as

$$h_v^{(k+1)} = U\left(h_v^{(k)}, AGG_{u \in N(v)} M\left(h_u^{(k)}, e_{uv}\right)\right) \qquad (1)$$

where M builds a message from the neighbour state and the bond feature e_{uv}, AGG is a permutation-invariant aggregator and U is the update function. After k layers a readout pools atom states into one molecule vector, which an MLP head maps to the twelve logits. We instantiate three architectures and two controls, summarised in Table II.

GCN applies four graph convolution layers with degree-normalised neighbour averaging, residual connections, batch normalisation and dropout, followed by concatenated mean and max pooling. Bond features are not used, which makes it the simplest inductive bias we test.

GINE-MPNN uses four GIN layers whose messages add a learned projection of the bond features before an MLP update, with sum and mean pooling. Sum aggregation makes it the most expressive of the three in the Weisfeiler–Lehman sense.

AttentiveFP learns attention weights over neighbours and applies a gated recurrent readout over two time steps, so that different atoms contribute unequally to the molecule vector.

FP-MLP is the key control: a multitask perceptron over exactly the same 2,048-bit Morgan fingerprint and 217 RDKit descriptors used by the classical baselines. Were the graph networks to outperform those baselines, this control would reveal whether the gain came from the graph representation or merely from deep multitask learning on shared features.

Hybrid concatenates the GINE molecule embedding with a regularised fingerprint branch and trains both end to end, as described in Section IV-F.

TABLE II
Deep Models Compared (Hidden Size 256)

Model	Representation	Depth / readout	Params
FP-MLP	fingerprint + descriptors	2262→1024→256	2.59 M
GCN	graph, no bond features	4 layers, mean ‖ max	0.41 M
GINE-MPNN	graph + bond features	4 layers, sum ‖ mean	0.69 M
AttentiveFP	graph + bond features	3 layers, 2 readout steps	1.93 M
Hybrid	graph + fingerprint	GINE ‖ FP branch	1.47 M

D. Masked, class-weighted loss

Let y_{it} be the label of molecule i for endpoint t, let $m_{it} = 1$ when that label is observed and 0 otherwise, let z_{it} be the corresponding logit, σ the logistic function, and w_t the ratio of negatives to positives for endpoint t in the training partition. The objective is

$$L = \left(\Sigma_{i,t}\, m_{it}\, \ell_{it}\right) / \left(\Sigma_{i,t}\, m_{it}\right), \qquad \ell_{it} = -\, w_t\, y_{it} \log \sigma(z_{it}) - (1 - y_{it}) \log(1 - \sigma(z_{it})) \qquad (2)$$

Missing entries therefore contribute neither loss nor gradient, and normalising by the number of observed labels keeps the loss scale independent of how many labels happen to be absent from a mini-batch. The endpoint-specific weight w_t reaches roughly 34

for the rarest endpoint, counteracting the tendency of an unweighted model to predict the majority class everywhere.

E. Training setup

All networks use AdamW (learning rate 10^{-3}, weight decay 10^{-5}), batch size 128, gradient clipping at 5, learning-rate halving after ten epochs without improvement, and early stopping with patience 20 over at most 100 epochs on mean validation ROC-AUC, restoring the best weights. Experiments ran on an eight-core Apple M3 CPU; a single graph network takes 4–15 minutes per split and the complete pipeline about two hours.

F. Combining the two representations

End-to-end hybrid. A naive concatenation of the graph and fingerprint embeddings performed worse than either component alone. The fingerprint branch fits the training set within two to four epochs, whereas the graph branch needs thirty to ninety, so early stopping halted training while the graph branch was still undertrained. The final hybrid applies input dropout of 0.5 and a narrow projection to the fingerprint branch, a stronger weight decay of 10^{-2} on that branch only, and *modality dropout*: during training the entire fingerprint embedding is zeroed for a random half of the molecules, forcing the prediction head to use the graph embedding as well. On the first scaffold split this raised validation ROC-AUC from 0.784 to 0.829 and moved the best epoch from 31 to 60.

Late-fusion ensemble. For each split, the graph model with the highest *validation* ROC-AUC is paired with the random forest. Per endpoint, the two models' scores are rank-normalised and mixed as

$$s = w \cdot r_{GNN} + (1 - w) \cdot r_{RF}, \quad w \in \{0, 0.25, 0.5, 0.75, 1\} \tag{3}$$

with w selected on the validation partition. Ranks make the differently calibrated outputs comparable, and ROC-AUC depends only on the ordering. Importantly, if the graph model carried no information beyond the fingerprint, validation would select $w = 0$ and the ensemble would collapse onto the random forest.

G. Evaluation protocol

For each endpoint, ROC-AUC and PR-AUC are computed on the test molecules that carry an observed label for that endpoint; missing labels are excluded from that endpoint's metric rather than imputed. Endpoint scores are macro-averaged and reported as mean ± standard deviation over the three split seeds. PR-AUC is reported alongside ROC-AUC because its chance level equals the prevalence, which makes it far more sensitive to how well the rare actives are ranked. Models are compared with two-sided paired Wilcoxon signed-rank tests over the 36 endpoint–seed pairs of test ROC-AUC [13].

V. RESULTS

A. Overall comparison

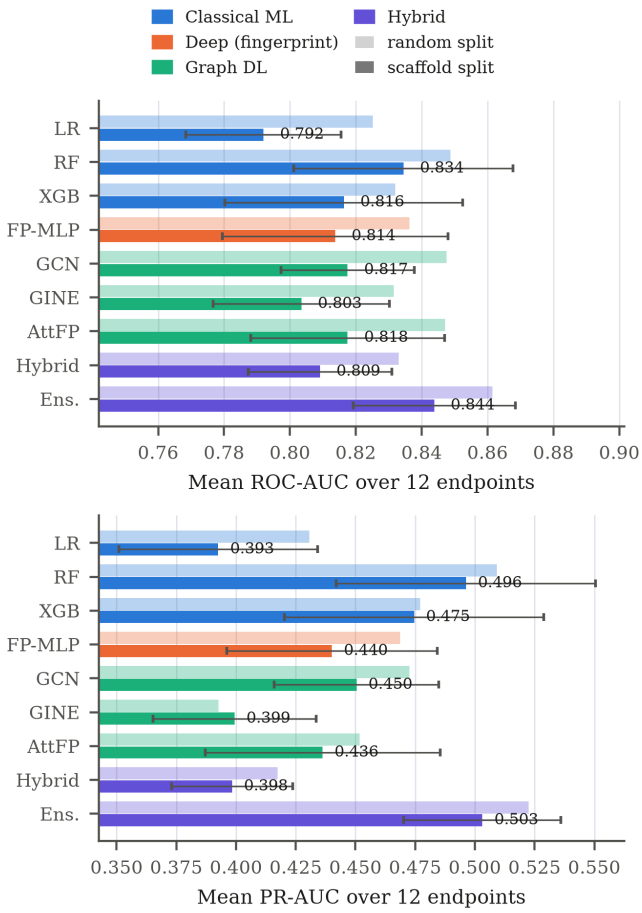


Fig. 2. Mean ROC-AUC (top) and PR-AUC (bottom) over the twelve endpoints. Solid bars are the scaffold split, with error bars showing the standard deviation over three seeds; pale bars behind them are the random split. Colour encodes model family.

TABLE III
Test Performance Averaged Over Twelve Endpoints (Mean ± SD Over Three Seeds). Gap = ROC-AUC(Random) − ROC-AUC(Scaffold)

Model	ROC-AUC (scaffold)	PR-AUC (scaffold)	ROC-AUC (random)	Gap
Logistic regression	0.792 ± 0.024	0.393 ± 0.042	0.825	0.033
Random forest	**0.834 ± 0.033**	**0.496 ± 0.054**	**0.849**	**0.014**
XGBoost	0.816 ± 0.036	0.475 ± 0.054	0.832	0.016
XGBoost (FP only)	0.751 ± 0.017	0.394 ± 0.027	—	—
FP-MLP	0.814 ± 0.034	0.440 ± 0.044	0.836	0.023
GCN	0.817 ± 0.020	0.450 ± 0.034	0.847	0.030
GINE-MPNN	0.803 ± 0.027	0.399 ± 0.034	0.832	0.028
AttentiveFP	0.818 ± 0.029	0.436 ± 0.049	0.847	0.030
Hybrid	0.809 ± 0.022	0.398 ± 0.025	0.833	0.024
Ensemble RF+GNN	**0.844 ± 0.025**	**0.503 ± 0.033**	**0.861**	**0.018**

Table III and Fig. 2 give the headline comparison. On scaffold splits the three graph networks reach 0.803 (GINE-MPNN), 0.817 (GCN) and 0.818 (AttentiveFP) ROC-AUC. These sit close to gradient boosting (0.816) and to the fingerprint perceptron (0.814), but clearly below the random forest (0.834), which remains the strongest single model. The descriptor ablation is informative in the other direction: XGBoost restricted to fingerprint bits falls to 0.751, so the global physicochemical

descriptors contribute more to the classical pipeline than any architectural choice contributes to the deep one. The late-fusion ensemble is the best model overall on both metrics (0.844 ROC-AUC, 0.503 PR-AUC).

B. Statistical significance

TABLE IV
Paired Wilcoxon Signed-Rank Tests on Scaffold-Split ROC-AUC Over 36 Endpoint–Seed Pairs

A	B	Δ ROC-AUC	Wins A / B	*p*	Sig.
RF	GCN	+0.0169	25 / 11	0.004	yes
RF	AttFP	+0.0169	31 / 5	< 0.001	yes
RF	GINE	+0.0310	30 / 6	< 0.001	yes
RF	FP-MLP	+0.0208	26 / 10	< 0.001	yes
RF	Hybrid	+0.0254	27 / 9	< 0.001	yes
XGB	GCN	-0.0012	17 / 19	0.810	no
XGB	AttFP	-0.0012	17 / 19	0.871	no
FP-MLP	GCN	-0.0038	17 / 19	0.560	no
GCN	GINE	+0.0140	24 / 12	0.007	yes
GCN	AttFP	-0.0000	18 / 18	0.981	no
AttFP	Hybrid	+0.0084	23 / 13	0.088	no
Ensemble	RF	+0.0093	26 / 10	0.001	yes

Table IV confirms the pattern. The random forest is significantly better than every graph network (+0.017 against GCN, $p = 0.004$) and than the fingerprint perceptron. By contrast, the graph networks cannot be distinguished from gradient boosting (-0.0012, $p = 0.81$) or from the perceptron, so deep learning on these data neither helps nor hurts relative to a well-tuned classical model. Two further comparisons are notable. The more expressive bond-aware GINE-MPNN is significantly *worse* than the plain GCN (-0.014, $p = 0.007$), indicating that additional capacity overfits at this data scale. The ensemble improves on the random forest in 26 of 36 pairs ($p = 0.001$), the clearest evidence that the graph view carries complementary signal.

C. Endpoint-wise behaviour

Fig. 4 breaks the comparison down by endpoint. GCN attains the best score of any model on NR-Aromatase (0.804 against 0.750 for the random forest), and AttentiveFP is the strongest single model on NR-PPAR-gamma; both are endpoints where the classical pipeline is comparatively weak. Conversely, gradient boosting is best on SR-ARE and SR-MMP, the two endpoints with the most positives. PR-AUC remains far below ROC-AUC on the rarest endpoints — SR-ATAD5 reaches 0.85 ROC-AUC but only 0.23 PR-AUC — confirming that ranking the few actives at the very top of a screening list is still difficult even when overall discrimination looks good.

D. Training dynamics

Fig. 3 explains why joint training of the two views is awkward. The fingerprint perceptron drives its training loss below 0.1 within roughly twenty epochs while its validation ROC-AUC peaks at epoch two or three: it memorises the training set almost immediately. The graph networks reduce their loss slowly and continue to improve on validation for thirty to ninety epochs. A single early-stopping criterion applied to a network containing both branches is therefore premature for one branch and far too late for the other, which is precisely the failure that the modality-dropout regulariser of Section IV-F was designed to address.

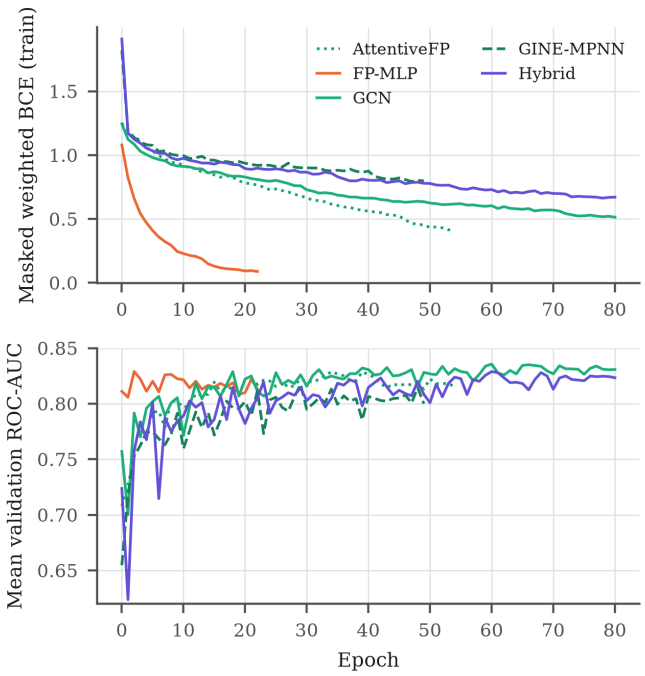


Fig. 3. Training loss (top) and mean validation ROC-AUC (bottom) on the first scaffold split. The fingerprint perceptron converges within roughly twenty epochs; the graph networks need thirty to ninety and are still improving when the perceptron has long since overfitted.

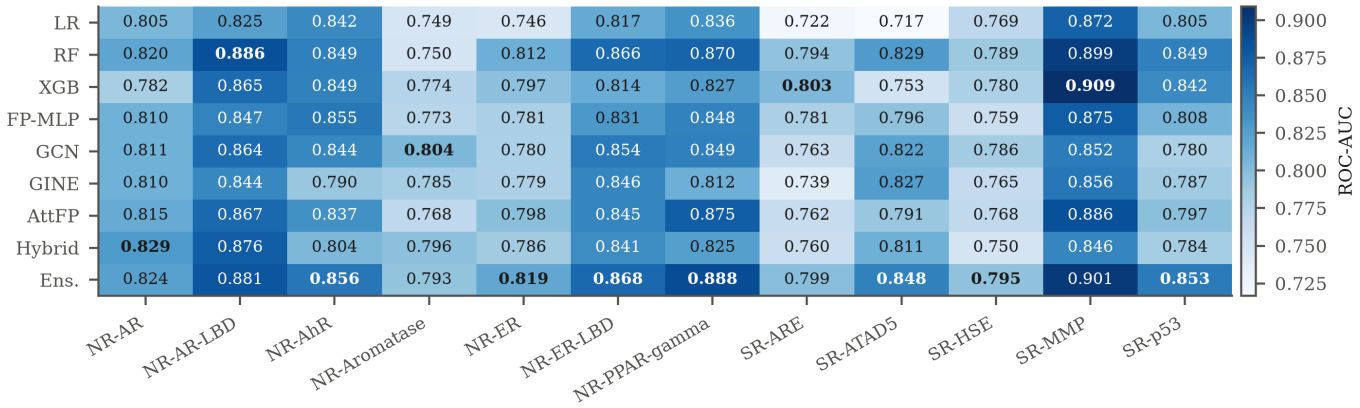


Fig. 4. Endpoint-wise test ROC-AUC on the scaffold split, averaged over three seeds. Bold entries mark the best model for each endpoint. No single representation dominates: graph networks win NR-Aromatase and NR-PPAR-gamma, gradient boosting wins the two largest stress-response assays, and the ensemble wins the majority.

VI. SCAFFOLD GENERALISATION

A. Random versus scaffold splits

Every model scores lower on scaffold splits than on random splits, confirming that random splitting overestimates performance on genuinely new chemistry. Part of this optimism is near-duplicate leakage: 7.1% of random-split test molecules have a fingerprint identical to a training molecule — mostly stereoisomers, which are distinct compounds but share a non-chiral ECFP — against 0.4% under scaffold splitting. The size of the gap differs systematically between families (Fig. 5). The tree ensembles lose 0.014 (random forest) and 0.016 (XGBoost), whereas the graph networks lose 0.028 to 0.030, roughly twice as much.

This is the most consequential finding of the phase. On *random* splits GCN (0.847) and AttentiveFP (0.847) are statistically level with the random forest (0.849); their apparent disadvantage emerges only once the test chemistry is new. A learned representation is fitted to the training distribution, so novel scaffolds present atom environments that the message functions have rarely seen, whereas fixed fingerprints and global descriptors are not learned and therefore cannot overfit to the training chemotypes in the same way.

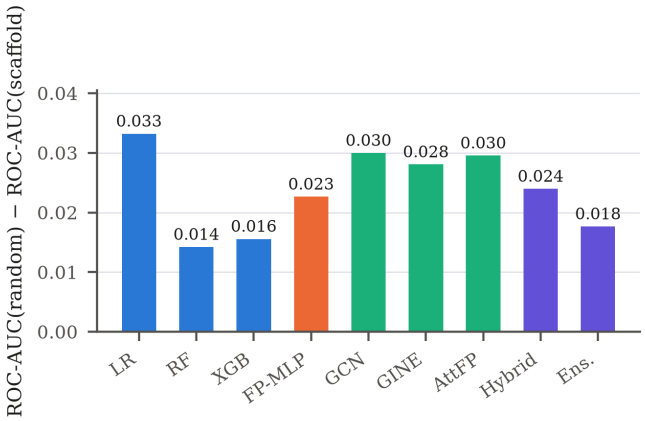


Fig. 5. ROC-AUC lost when moving from a random to a scaffold split, averaged over endpoints. Lower is better. The graph models (green) lose about twice as much as the tree ensembles (blue).

B. Distance to the training set

To make the effect concrete, each test molecule is binned by the maximum Tanimoto similarity between its Morgan fingerprint and those of all training molecules, pooling the test sets of all six splits (Fig. 6). Accuracy declines monotonically with novelty for every model: from 0.85–0.88 ROC-AUC for molecules with a close training analogue (similarity at least 0.7) down to 0.64–0.72 below 0.3. In that hardest bin the random forest (0.718) and GINE-MPNN (0.706) degrade least among single models, XGBoost degrades most (0.637), and the ensemble remains best (0.723). For deployment this argues for an explicit applicability-domain check: predictions for compounds whose nearest training neighbour falls below a similarity of roughly 0.3 to 0.4 should be flagged as unreliable rather than reported as scores.

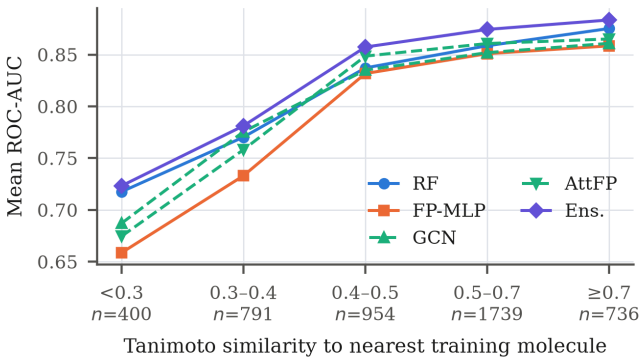


Fig. 6. Mean ROC-AUC against each test molecule's Tanimoto similarity to its nearest training molecule, with the test sets of all six splits pooled. Every model degrades sharply on genuinely novel chemistry.

VII. DISCUSSION

A. Why learned representations do not win here

Three factors explain the result, and they reinforce one another. First, Tox21 provides only about 6,100 training molecules and between 180 and 900 actives per endpoint, which is small for representation learning; a Morgan fingerprint is itself essentially a Weisfeiler–Lehman style hash of the same neighbourhoods a message passing network aggregates, so the network must learn from scarce data something the fingerprint already encodes for free. Second, the baseline is unusually strong because it is not a fingerprint alone: the descriptor ablation shows that global properties such as lipophilicity, molecular size and polar surface area carry a large part of the signal, and a graph network must infer these from atom-level features. Third, scaffold splitting penalises learned representations more than fixed ones, as Section VI showed. Our conclusion is consistent with MoleculeNet [2] and Chemprop [12], where graph models pull clearly ahead mainly on larger datasets, with added descriptors, or after self-supervised pre-training.

The FP-MLP control rules out the most obvious alternative explanation. Trained on identical features, it reaches 0.814 ROC-AUC, statistically indistinguishable from the graph networks and below the random forest. Deep multitask learning by itself therefore neither explains nor rescues the comparison: the architecture family is not the limiting factor, the data scale is.

B. Complementarity of the two views

Although graph networks do not win on average, they are not redundant. They are the best single models on two endpoints, and when the validation set is free to choose the mixing weight of equation (3) it assigns on average 0.46 of the weight to the graph model rather than collapsing to zero. The resulting ensemble improves significantly on the random forest and holds up best of all models on the most novel chemistry. The two representations evidently make different errors: engineered features capture known substructures and bulk physicochemical behaviour, while the learned representation adapts to endpoint-specific patterns. Combining them is more effective than replacing one with the other.

C. Expressiveness is not the bottleneck

A notable negative result is that the most expressive architecture performs worst. GINE-MPNN uses sum aggregation and bond-conditioned messages, making it strictly more discriminative than GCN in the Weisfeiler–Lehman sense, yet it falls significantly below GCN and has the lowest PR-AUC of the graph models. With roughly 6,100 training molecules the additional capacity is spent fitting training chemotypes. This supports the view that, in low-data molecular property prediction, regularisation and inductive bias matter more than theoretical expressive power.

D. Reliability of the comparison

Split-to-split variation (standard deviations of 0.02–0.04 ROC-AUC) exceeds most differences between models, so conclusions drawn from a single split could easily be reversed. Pairing comparisons by endpoint and seed removes this shared difficulty, and we claim only the differences that are significant in Table IV. Because endpoints are correlated and seeds share molecules, the p-values should be read as approximate; nevertheless the principal random-forest-versus-graph results ($p \leq 0.004$) survive a Bonferroni correction over the planned comparisons.

VIII. LIMITATIONS AND FUTURE WORK

Unequal tuning. The classical baselines received a per-endpoint grid search, whereas each network used a single fixed configuration. A systematic search over depth, width, dropout and learning rate could narrow the gap, although the consistency of the result across three architectures suggests it would not reverse it.

No pre-training. Self-supervised pre-training of the graph encoder on large unlabelled corpora is the most promising route to stronger learned representations in this regime, and is the obvious next experiment.

Calibration and uncertainty. Class-weighted losses optimise ranking rather than probabilities, so the outputs would need Platt or isotonic calibration before being used as risk estimates, and deep ensembling could supply uncertainty intervals.

Stereochemistry and split strictness. ECFP ignores chirality and our graph models see only a chirality tag, so stereoisomers carrying different labels cannot be separated reliably. Scaffold splitting is also imperfect, since different scaffolds can still be chemically similar; cluster-based or temporal splits would test extrapolation more strictly.

IX. CONCLUSION

We compared engineered chemical fingerprints with learned molecular-graph representations for predicting twelve Tox21 toxicity endpoints under a leakage-controlled scaffold-split protocol. Three message passing architectures trained as multitask models with a masked, class-weighted loss reach 0.803–0.818 ROC-AUC on unseen chemical families. They are statistically indistinguishable from gradient boosting and from a deep perceptron over the same engineered features, and significantly below a random forest over fingerprints and descriptors (0.834). Graph networks also generalise less well from random to scaffold splits, losing about twice as much accuracy as the tree ensembles. The two views are nonetheless complementary: a validation-weighted late fusion of a random forest and a graph network is the best model overall (0.844 ROC-AUC, 0.503 PR-AUC) and degrades least on novel chemistry. For datasets of this size we therefore recommend combining engineered and learned repres-

entations rather than substituting one for the other, and evaluating under scaffold splits, where differences in generalisation become visible at all.

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